



Design of multi-modal feedback channel of human–robot cognitive interface for teleoperation in manufacturing

Chen Zheng^{1,2} · Kangning Wang¹ · Shiqi Gao¹ · Yang Yu¹ · Zhanxi Wang¹ · Yunlong Tang^{3,4} 

Received: 13 March 2024 / Accepted: 21 June 2024 / Published online: 9 July 2024
© The Author(s) 2024

Abstract

Teleoperation, which is a specific mode of human–robot collaboration enabling a human operator to provide instructions and monitor the actions of the robot remotely, has proved beneficial for application to hazardous and unstructured manufacturing environments. Despite the design of a command channel from human operators to robots, most existing studies on teleoperation fail to focus on the design of the feedback channel from the robot to the human operator, which plays a crucial role in reducing the cognitive load, particularly in precise and concentrated manufacturing tasks. This paper focuses on designing a feedback channel for the cognitive interface between a human operator and a robot considering human cognition. Current studies on human–robot cognitive interfaces in robot teleoperation are extensively surveyed. Further, the modalities of human cognition that foster understanding and transparency during teleoperation are identified. In addition, the human–robot cognitive interface, which utilizes the proposed multi-modal feedback channel, is developed on a teleoperated robotic grasping system as a case study. Finally, a series of experiments based on different modal feedback channels are conducted to demonstrate the effectiveness of enhancing the performance of the teleoperated grasping of fragile products and reducing the cognitive load via the objective aspects of experimental results and the subjective aspects of operator feedback.

Keywords Robot teleoperation · Human–robot collaboration · Multi-modal feedback channel · Human–robot cognitive interface · Cognitive load

Introduction

Robotic manufacturing systems have become increasingly popular in global manufacturing because of their advantages such as high speed, accuracy, and precision (Zhao et al., 2023a, 2023b). However, given the intricate nature of the manufacturing environment, the concept of combining

human intelligence and dexterity with the precision and physical capabilities of robots has become the most promising approach to handle complex manufacturing tasks involving factors such as product variations, customization, quality control, and nonroutine problem solving (Arents & Greitans, 2022; Zhao et al., 2022).

Robot teleoperation is a specific mode of human–robot collaboration that enables a human operator to provide instructions and monitor the actions of a robot in dangerous and unstructured environments remotely. Robot teleoperation is particularly valuable for minimizing risks and ensuring the safety of human operators (Fang et al., 2019). Further, robot teleoperation enables human operators to perform complex and precise manufacturing tasks (e.g., complex assembly, welding, part replacement, quality control, and defect identification) using a cognitive interface to enhance flexibility during the manufacturing process (Zhao et al., 2023a, 2023b; Zhou et al., 2020).

The cognitive interface between the robot and human operator is of paramount importance in the aforementioned

✉ Yunlong Tang
yunlong.tang1@monash.edu

¹ School of Mechanical Engineering, Northwestern Polytechnical University, 127 West Youyi Road, Xi'an 710072, Shaanxi, P. R. China

² Guangdong Provincial Key Laboratory of Advanced Welding Technology for Ships, CSSC Huangpu Wenchong Shipbuilding Company Limited, Guangzhou, P. R. China

³ Mechanical and Aerospace Engineering Department, Monash University, Melbourne 3168, Australia

⁴ Materials Science and Engineering Department, Monash University, Melbourne 3168, Australia

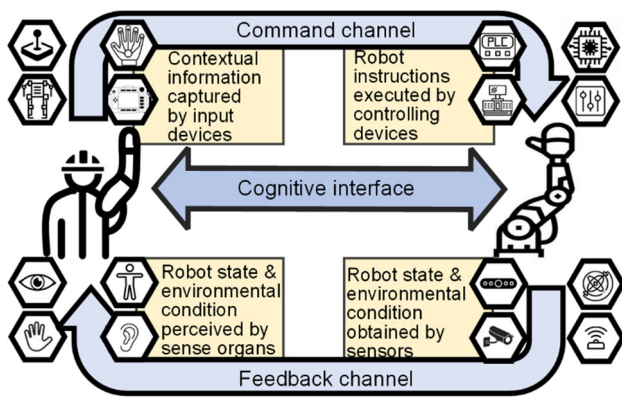


Fig. 1 Cognitive interface between human operator and robot

teleoperated robotic manufacturing process because it helps facilitate effective communication, collaboration, and information exchange between the human operator and robot, which offers the human operator a more intuitive way to control the robot remotely, ultimately improving manufacturing performance (Chacón et al., 2021). Figure 1 shows that the cognitive interface not only enables the robot to perceive and understand the contextual information provided by the human operator (i.e., the command channel from the human operator to the robot); however, it enables the robot to not only interpret the human operators' high-level commands considering task-specific constraints and adapt its behavior accordingly, but also facilitate the provision of cognitive assistance and feedback from the robot to the human operator (i.e., feedback channel from the human operator to the robot), helping the human operator monitor the state of the robot, understand the current condition of the environment, and make appropriate adjustments or interventions.

Multi-modal data integration approaches are being widely used for constructing command channels that integrate data captured from different input devices or sensors to enable robots better understand the actions and intentions of human operators (Kong et al., 2019). Compared to the multi-modal data integration approaches employed for controlling robots, most feedback channels in the current manufacturing domain only adopt visual feedback. Although visual feedback provides the human operator with one of the most direct approaches to gain situational awareness of the robot and manufacturing environment, evaluate the state of the workspace, and identify the unstructured environments and potential hazards by observing the live video feed or graphical representation of the surroundings of the robot, such a single-modal interaction approach can potentially increase the cognitive load of the human operator (Darvish et al., 2023; Huang et al., 2022). Given this limited sensory information, human operators must exert more mental effort to compensate for the lack of feedback from other modalities, which can lead to higher mental fatigue, reduced attention,

and decreased performance, particularly in manufacturing tasks that demand a high level of precision and concentration (Jeong et al., 2024). Although multi-modal information can enhance cognitive processes in the feedback channel, it does not guarantee better cognitive performance (Su et al., 2023). Redundant modalities and poorly designed or poorly integrated multi-modal information can still lead to confusion, distraction, or cognitive conflict among human operators. Therefore, the selection, integration, and coordination of different modalities for the feedback channel of the cognitive interface between the human operator and robot must be considered carefully, which ensures the human operators monitor the robot's state, understand the current condition of the environment, and make appropriate decisions or interventions.

This study focuses on the design of a feedback channel in the cognitive interface between the human operator and the robot from the perspective of human cognition. The paper is organized as follows: “[Literature review](#)” Section analyzes the limitations of the human–robot interface in robot teleoperation based on an in-depth survey of the evolving relationship between human operators and robots. From the perspective of human cognition, “[Design of multi-modal feedback channel of human–robot cognitive interface](#)” Section presents the design of the feedback channel in the cognitive interface to decrease the cognitive load of the human operator during the teleoperated robotic manufacturing process. The channel design includes the selection, integration, and coordination of multi-modal feedback information. “[Implementation of feedback channel in teleoperated robotic grasping system](#)” Section presents a case study based on a teleoperated robotic grasping system in which the proposed multi-modal feedback channel is implemented. The experiments are based on objective and subjective assessments, demonstrating the effectiveness of reducing the cognitive load of the human operator while remotely grasping fragile products by implementing the cognitive interface proposed in “[Experiment and evaluation](#)” Section. “[Discussion](#)” Section provides an in-depth discussion of the cognitive interface from the perspective of cognitive load. Finally, the conclusions are presented in “[Conclusion](#)” Section.

Literature review

Human–robot collaboration and cognitive interface

Figure 2 shows the different integration levels between human operators and robots. In 1961, American mechanical engineers Joseph F. Engleberger and George Devol pioneered the creation of UNIMATE (Braschetti et al., 2023), widely recognized as the world's first commercial industrial robot. Initially designed to work autonomously and be isolated from

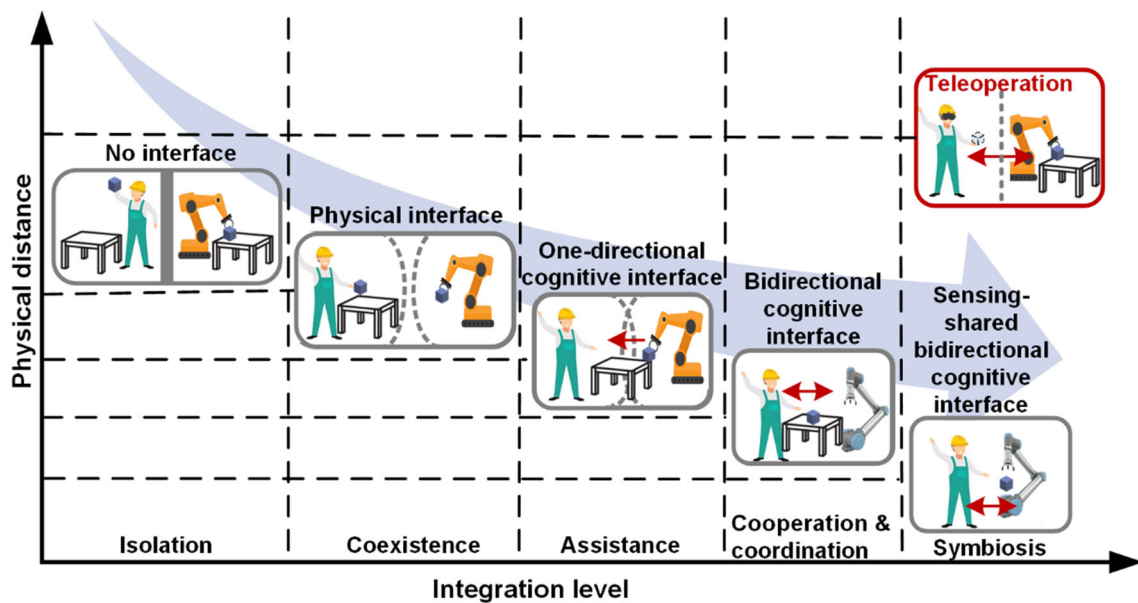


Fig. 2 Integration levels for human–robot collaboration

human operators, UNIMATE could autonomously perform potentially hazardous, strenuous, and monotonous tasks. Since then, such isolated industrial robots have been widely used in the manufacturing domain, including welding, material handling, and assembly work, because they can perform various tasks based on preprogrammed instructions throughout the manufacturing process (Ji & Wang, 2019; Zhou et al., 2024). The term “isolation” describes the relationship between human operators and robots at that time: Robots were placed in isolated environments for distinct workspaces and tasks and operated independently without direct interaction, minimizing the potential safety risks but limiting collaboration (Javaid et al., 2021).

In the late 1990s, the desire for more flexible and adaptive manufacturing processes led to the human–robot collaborative manufacturing mode (Gervasi et al., 2020) that requires robots to work besides human operators in shared workspaces. This coexistence between human operators and robots marks the transition towards greater human–robot collaboration. At the base level, a coexistence relationship requires humans and robots to coexist in the same workspace and perform separate tasks without direct interaction (Liu et al., 2024). That is, they share the physical space but work independently, often with physical barriers or safety measures to ensure human safety. As collaboration advances, robots are expected to start assisting human operators perform specific tasks (Kolbeinsson et al., 2019). Here, the relationship becomes one-directional, wherein robots provide support based on predefined instructions or commands from the human operators. As the technology continues to evolve, cooperation and coordination will signify higher

levels of collaboration, with humans and robots working together to attain a shared objective. (Keshvarparast et al., 2023). At this level of cooperation, humans and robots share information and communicate bidirectionally, with each entity contributing their unique capabilities to achieve mutual objectives. At the level of coordination, both parties communicate actively and adapt their actions based on the status and progress of the other, thereby involving a strong sense of teamwork and dynamic coordination between them (Castro et al., 2021; Nguyen Ngoc et al., 2022). Symbiosis is currently considered the most advanced level of collaboration between humans and robots, where they seamlessly combine their strengths and compensate for their weaknesses to focus on more complex and cognitively intensive activities by sharing their sensing through the cognitive interface. Further, robot teleoperation, a specific type of human–robot symbiotic collaboration, was the most productive collaboration in hazardous and unstructured manufacturing environments, and it helps ensure the safety of operators to the maximum extent (Li et al., 2023). Table 1 lists the specific attributes of the different integration levels for human–robot collaboration in different periods to further clarify the distinctions between them.

Robot teleoperation in the manufacturing process

The proposition of robot teleoperation can be traced back to the 1940s, when researchers had only started exploring approaches for enabling humans to interact with robots located in environments that were either inaccessible or hazardous to humans (Moniruzzaman et al., 2022). Given the

Table 1 Specific attributes of human–robot collaborations in different periods

	Interaction		Same workspace	Precise task	Dangerous task
	Human to robot	Robot to human			
Isolation	–	–	–	–	✓
Coexistence	–	–	–	–	–
Assistance	–	✓	✓	–	–
Cooperation & coordination	✓	✓	✓	–	–
Symbiosis	✓	✓	✓	✓	–
Teleoperation	✓	✓	–	✓	✓

recent developments in the sensor technologies, current teleoperated robotic manufacturing processes offer human–robot symbiotic collaboration, wherein the actions and decisions of humans directly influence the behavior of the robot through the cognitive interface, and vice versa (Almeida et al., 2020). Therefore, the cognitive interface in a teleoperated robotic system enables bidirectional communication, through which the robot performs physical tasks based on guidance provided by the human operator (i.e., command channel), while the human operator receives sensory information from the sensors of the robot (i.e., feedback channel), such as video feeds, haptic feedback, and other relevant data, to make informed decisions during teleoperation.

Currently, various approaches and relevant techniques have been developed to construct the command channel of a teleoperated robotic system to enable human operators to send command signals with different modalities to robots (Pan et al., 2021). Based on the interaction mode, the current command channel can be divided into natural-language-based and gesture-based command channels. Natural language-based command channels enable human operators issue commands using voice by integrating speech recognition approaches (Liu & Zhang, 2019). Various natural language processing algorithms, such as neural machine translation (Ehsan et al., 2018), hidden Markov model (HMM) (Mustafa et al., 2019), and Mel-Frequency Cepstrum Coefficients (MFCC) (Anggraeni et al., 2018), have been used to translate the speech of a human operator into an executable set of actions for the robot. Gesture-based command channels offer an intuitive method for operators to control robots that use gestures or body movements of a human operator to command robots, thereby aiming to simplify control. Conventional input devices include keyboard, mouse, joystick, etc. (Mahmud et al., 2020). New technologies, such as virtual reality (VR), augmented reality (AR), mixed reality (MR), and various artificial intelligence algorithms, has led to the development of different input devices or sensors that can enable robots to better understand the actions and intentions of human operators. For example, Du et al. (2012) adopted a teleoperated control method that tracks

the index finger and thumb and processes depth images from Kinect. Chamorro et al. (2021) developed a gesture recognition system for estimating the pose of a human operator using lidar scanning results. Kou et al. (2023) used an electromyographic (EMG) signal from the muscles of human hand motions for constructing a command channel. Qiu et al. (2017) analyzed electroencephalography (EEG) signals of human operators and presented a teleoperation control approach that enabled a robot to complete tasks based on the EEG reading of human operator. Yu et al. (2014) presented a gaze-based teleoperation approach that tracked the gaze of the operator.

As to the application of the feedback channel, most of the current teleoperated robotic systems adopt only live videos as the feedback channel for remotely monitoring the actions of a robot operating in hazardous and unstructured manufacturing domains (Bejczy et al., 2020). Traditionally, human operators received visual information from industrial cameras fixed on robots, walls, or adjacent structures in the manufacturing environment (Bandala et al., 2019). The feedback view of an industrial camera could only be accepted by a computer screen, which is not natural, and it can be visualized by human operators to complete remote manufacturing tasks. Given the support of emerging technologies, various novel approaches were developed to construct vision-based feedback channels. Naceri et al. (2019) utilized an RGB camera and point-cloud data for constructing visual feedback that can offer human operators high-fidelity graphical rendering on the VR device. Solanes et al. (2020) developed an AR-based visual feedback channel that utilized a depth camera mounted on an industrial robot to capture environmental information and transmit it AR goggles. Su et al. (2022) adopted MR technology based on an HTC VIVE device to help human operators perceive immersive 3D/2D views of the manufacturing environment. Although the aforementioned vision-based feedback channels are the most direct way for human operators to obtain the situational awareness of the robot and manufacturing environment, such a single-modal approach potentially burdens the human operators. Zhu et al.

(2023) identified the limitations of the proposed teleoperated grasping control framework based on visual feedback for complex grasping tasks. They claimed that integrating alternative feedback channels would help enhance the overall system performance. Yang et al. (2023) demonstrated this viewpoint by considering the cognitive load of operators when using an AR programming interface based on visual feedback. Although the interface completed complex tasks effectively and efficiently, some operators claimed that it was not sufficiently intuitive to operate based solely on visual information. The limited sensory information requires human operators to exert additional mental effort to compensate for the lack of feedback from other modalities, and therefore, this can result in higher mental fatigue and decreased performance, particularly in manufacturing tasks that demand a high level of precision and concentration.

Some researchers have started considering other types of sensory information and combining them with visual information when designing feedback channels. For example, Yashin et al. (2019) utilized vibrotactile feedback based on visual feedback to enhance the interaction of human operators, who can perceive the products in remote environments. Ding et al. (2022) employed auditory feedback for providing information relevant to the manufacturing environment to assist with visual feedback and enhance the operational efficiency of hydraulic manipulators. Further, although some researchers studied multi-modal human–robot interactions that integrate vision, auditory, haptics, or physiology, teleoperated robotic manufacturing systems intertwined with multi-modal feedback channels remain underexplored (Wang et al., 2023). Random selection and poor integration of multi-modal sensory information can result in confusion, distraction, or cognitive conflict for the human operator. Consequently, the application of multi-modal feedback channels remains an urgent problem that needs to be addressed in the teleoperated manufacturing domain.

Design of multi-modal feedback channel of human–robot cognitive interface

Design requirements of teleoperated robotic grasping system

An industrial partner required the authors' research team to improve the design of the feedback channel of a teleoperated robotic grasping system for enabling a human operator to remotely control a robotic arm to perform grasping tasks for an extremely fragile product in the aerospace domain. Previously, the fragile products were cylindrical and made of combustible power, requiring human operators to grasp and transport the combustible product by hand. Considering the dangerous and unstructured manufacturing environment, this

company developed a teleoperated robotic grasping system to replace human operators in performing the grasping tasks. However, the current teleoperated robotic grasping system only adopts live video as visual feedback (i.e., a single-modal feedback channel) to assess the current state of the workspace, which cannot guarantee the precision and concentration of the complex manufacturing task. Owing to the limited information regarding the visual feedback, human operators need to exert more mental effort to compensate for the lack of feedback from other modalities, which can lead to a high cognitive load. Therefore, this study designs a teleoperated robotic grasping system as follows: (1) Designing an immersive and intuitive interface for human operators to understand the remote environment. (2) Human operators complete grasping tasks without destroying fragile products. (3) The design of the human–robot interface considers the cognitive load to ensure the efficiency of grasping tasks.

Cognitive loads in teleoperated robotic manufacturing process

Considering the limitations of the current feedback channels, a multi-modal design solution for the feedback channel from the perspective of human cognition to decrease the human operator's cognitive load is proposed in this paper.

Cognitive load, a concept in cognitive psychology, describes the mental effort or cognitive resources needed to complete tasks or acquire new information. This concept was first introduced by Sweller et al. (2019), whose work on cognitive load theory (CLT) acted as the fundamental framework for instructional design and presented the implications of presenting information effectively to learners, designing user interfaces, and optimizing human performance in complex tasks. According to the CLT, cognitive load can be categorized into three types based on the sources: intrinsic, extraneous, and germane cognitive load.

Intrinsic cognitive load arises from the inherent difficulty of the task determined by the complexity of the information being processed and the interactivity between the different elements of the task. Extraneous cognitive load is imposed by irrelevant elements that are not directly related to the task at hand, which hinder task performance by diverting cognitive resources. Germane cognitive load involves cognitive load that contributes to meaningful learning, schema construction, and problem solving, which is associated with building mental models, organizing information, and integrating new knowledge with existing knowledge.

The different cognitive loads during the teleoperated robotic manufacturing process must be clarified to reduce the cognitive load of a human operator. Therefore, a series of interviews were conducted with human operators, including two novices with less than 2 months of experience and two experts with more than 5 years of experience. This

Table 2 Three types of cognitive loads in teleoperated robotic manufacturing process

Types of cognitive load	Cognitive load problems of teleoperated manufacturing task
Intrinsic cognitive load	Coordinating the movements of a multi-limbed robot Interpretation of sensor data from various sources Inherent complexity of the teleoperated manufacturing task
Extraneous cognitive load	Limited sensory information in the single-modal interaction mode Poorly designed multi-modal feedback that includes irrelevant or excessive information Information from multiple modalities that are not in perfect synchronization
Germane cognitive load	Integration of the feedback from the robot and the environment with the existing knowledge and mental models of the operator Teleoperated manufacturing tasks involving learning and acquiring new skills

semi-structured interview was organized and centered on the following question: What are the difficulties in obtaining feedback from the robot? The interview results (Table 2) were divided into three groups based on different types of cognitive load.

Intrinsic cognitive load

The intrinsic cognitive load can be determined by the inherent complexity of the manufacturing task. In other words, manufacturing tasks with a higher intrinsic cognitive load require more mental effort and attention to be understood and solved. Compared to the experts, this type of cognitive load is relatively high for novices. According to interviews with novices, a human operator can experience a high intrinsic cognitive load when attempting to understand and predict a behavior of a robot in complex and dynamic environments. For example, coordinating the movements of a multi-limbed robot or interpreting sensor data from various sources can be cognitively demanding. Further, an intrinsic cognitive load can arise from the inherent complexity of teleoperated manufacturing tasks. For instance, solving intricate manipulation tasks, navigating cluttered environments, and making critical decisions based on incomplete or uncertain information can impose a high cognitive load on the operator.

Extraneous cognitive load

During the teleoperated robotic manufacturing process, the extraneous cognitive load is imposed by irrelevant information that distracts from the manufacturing task. Three examples at different levels can be used to describe this type of cognitive load. First, the feedback channel in most current teleoperated robotic manufacturing systems only adopts visual feedback to help the human operator gain situational awareness of the robot and manufacturing environment. However, given the limited sensory information, such single-modal interactions require human operators to exert more mental effort to compensate for the lack of feedback from other modalities, which can lead to a higher extraneous cognitive load, particularly in manufacturing tasks that demand a high level of precision and concentration. Second, some feedback channels begin to consider multi-modal information however, poorly designed multi-modal feedback that includes irrelevant or excessive information can increase the extraneous cognitive load. For example, presenting too many visual cues or auditory alerts not directly related to the operator's current manufacturing task can distract and overload cognitive resources, resulting in a high extraneous cognitive load. Third, although different types of sensors can be carefully selected to avoid irrelevant or excessive multi-modal information, information from multiple modalities that are not perfectly synchronized can still increase the extraneous cognitive load. For example, if the actions of the robot are presented first visually and then using auditory cues (even after a few milliseconds), such delayed feedback can influence the capture of robot movements and may reduce the concentration of the operator because the brain needs to allocate additional cognitive resources to reconcile the temporal or spatial disparities.

Germane cognitive load

Unlike the two aforementioned cognitive load, germane cognitive load is considered beneficial because it supports long-term retention and is expected to increase during the teleoperated robotic manufacturing process. On the one hand, the feedback channel plays a significant role for robot teleoperation because the germane cognitive load can arise from integrating the feedback from the robot and the environment with the existing knowledge and mental models of the operator. Therefore, constructing a new mental model that can be understood easily by operators and integrating it into the teleoperation process to form a coherent understanding of the teleoperation task can enhance germane cognitive load. On the other hand, interviews with operators revealed that increased experience with the robot led to greater fluency in performing teleoperated manufacturing tasks. Therefore, engaging in deliberate practice, reflecting on performance,

and integrating operational experiences into existing expertise can increase the germane cognitive load.

Design method of feedback channel

A design approach is proposed to offer human operators enhanced cognitive assistance to monitor the actions of the robot and make appropriate adjustments considering the three aforementioned types of cognitive loads during the teleoperated robotic manufacturing process. First, to reduce the intrinsic cognitive load, different sensory data with multiple robot modalities must be selected carefully for providing human operators with a more intuitive and direct approach for understanding the robot's actions and the manufacturing tasks' situation. Second, the extraneous cognitive load, which includes the selected sensory data with multiple modalities, should be interpreted as the cognitive resources of the human operators and well synchronized. Third, the germane cognitive load can be increased by integrating the perceived feedback to construct a new mental model for human operators. Moreover, engaging in deliberate practice and integrating new knowledge and skills with existing expertise can contribute to a germane cognitive load that supports learning and skill development.

Cognitive mapping-based sensory data selection

The combination of multi-modal feedback channels in current teleoperated robotic systems without careful selection confuses and distracts the operators. Sepp et al. (2019) explained this cognitive problem using the working memory model, which posits that a cognitive load occurs when human operators perform over-memory-capacity manufacturing tasks. Koelewijn et al. (2014) adopted the signal-to-noise ratio, which refers to the ratio of relevant information (i.e., signal) to irrelevant or distracting information (i.e., noise), to describe this phenomenon. The signal-to-noise ratio decreases with an increase in the number of cognitive cues, which makes it challenging for human operators to focus on relevant cues and leads to confusion. Consequently, Gestalt principles of perception suggest that humans naturally seek simplicity and organization in the information they receive (Baker et al., 2009). That is, human operators can efficiently process information without feeling overwhelmed by organizing cognitive cues into meaningful groups or patterns.

The mapping principle, which is an essential principle in human-machine interface (HMI) design, involves aligning the physical or spatial arrangement of controls with the corresponding functions or outcomes of the system to establish a clear and intuitive connection between the human and the machine within the command channel design. For example, a joystick that moves forward and backward to control the robot's movement in the same direction follows the principle

of natural mapping. Compared with the command channel, the design of the feedback channel requires aligning the sensory data of the robot's action and the workspace's situation with the cognitive abilities of human operators. In this study, the mapping principle is adopted to improve the proposed feedback channel and minimize the intrinsic cognitive load on human operators.

Human cognitive processes occur through five senses: vision, sound, taste, smell, and touch. Practical senses that enhance the perception of the action of a robot and reduce the cognitive load of human operators must be carefully selected considering the manufacturing context when designing the feedback channel of a teleoperated robotic grasping system. A human operator relies heavily on visual cues to dynamically obtain the manufacturing environment, recognize the product, and gather information about the spatial relationships between the product and its environment. Vision is thus the most dominant and highest-priority sense for humans, and therefore, achieving a mapping between human visual perception and teleoperated visual information is at the core of enhancing robot performance during teleoperation. Haptic perception is essential for exploring and manipulating fragile products in teleoperated grasping systems. Building a map between human haptic perception and the grasper in a teleoperated robotic grasping system can provide an intuitive force to the operators. Although the sense of sound is considered the second most important sense in terms of priority (as the brain processes auditory information quickly), it is less dominant than vision and touch in the teleoperated grasping systems. However, auditory cues can be used to complement visual and haptic perception for realizing the mapping between human auditory perception and potential environmental dangers when visual cues are limited or unavailable. The senses of taste (i.e., gustatory perception) and smell (i.e., olfactory perception) have a lower priority compared to that of the three senses mentioned above. Although they are important for the detection of certain environmental cues, they play a less central role in cognitive processing and decision making during the manufacturing process. Therefore, vision, sound, and touch are selected as the three senses that could be used to map the human operator and teleoperated robot by analyzing the human cognitive process and characteristics of the teleoperated robotic grasping system.

Multi-modal sensory data capturing

Visual perception for multi-modal feedback channel Most current teleoperated robots use a camera as a visual feedback channel for observing the robot and its manufacturing environment. Different views provided by the camera in teleoperated robots, which are critical for giving human operators an optimal view of both the robot and manufacturing environment, are presented below:

Overhead view Mounting a camera overhead can offer a bird's-eye view to monitor the entire manufacturing environment, including the robot and its surroundings. This allows the operator to obtain a comprehensive view of the spatial relationships between the robot, workpieces, and other equipment.

Side view A side-view camera mounted on a wall or adjacent structure can provide a lateral view of the robot's movements and the manufacturing environment. This view is beneficial for tasks that require a close coordination of the robot and its surroundings.

End-effector view Placing a camera on the robot's end effector provides the operator with a view from the robot's perspective. This view allows the human operator to determine the action of robot's end-effector.

Multiple camera views Using multiple cameras from different angles can offer a more comprehensive understanding of the environment based on the tasks' complexity and the robots' workspace. This approach enables the human operator to switch between camera views to optimize the situational awareness.

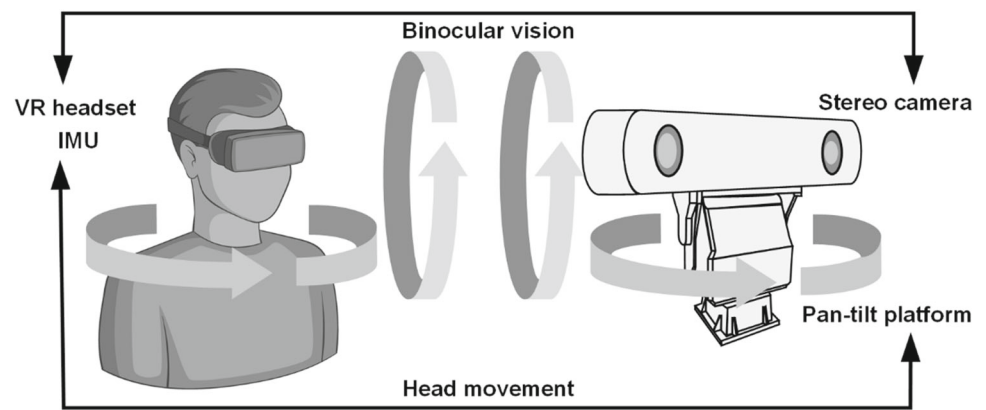
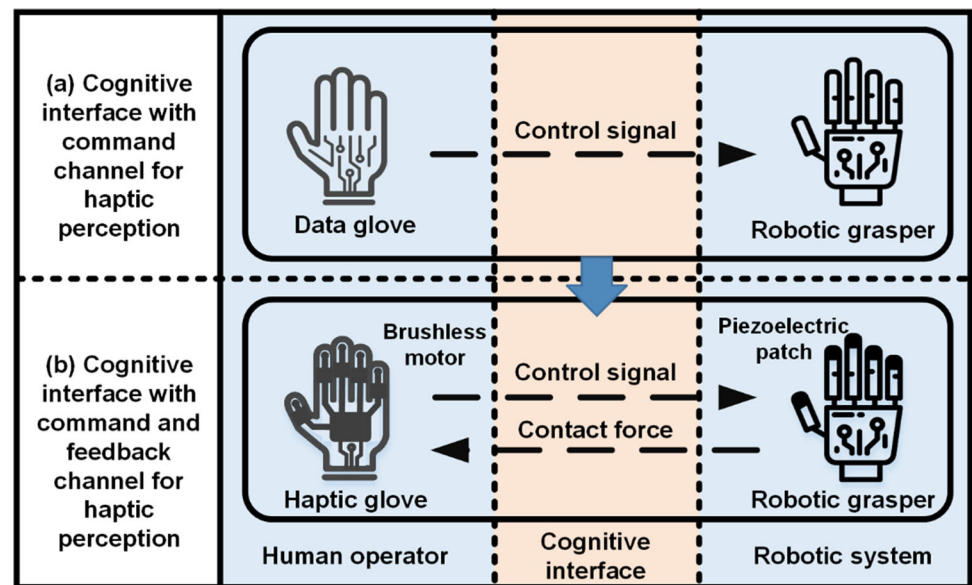
Both the overhead and side views require the camera to be mounted at a fixed position, which allows interactions between the robot and the environment. However, cameras with fixed positions have blind spots that limit the human operator's situational awareness and enhance the potential challenges. End-effector and multiple-camera views can be considered a mobile view that provides the operator with a view from different positions in the manufacturing environment. Although the end-effector view provides a more immersive and adaptable perspective for certain tasks compared with the overhead or side views, it does not provide a natural visual perception similar to that of the human operator, thereby leading to motion sickness. In addition, multiple camera views allow human operators to switch between different camera views, resulting in a highly extraneous cognitive load.

Providing a more immersive and adaptable view aligned with human perceptual and behavioral patterns, which makes teleoperated grasping tasks more natural, intuitive, and efficient, is necessary to overcome these limitations. The pan and tilt functionalities, which allow the camera to move horizontally (pan) and vertically (tilt) to replicate the natural movement of the head of the human operator, were designed and integrated into the feedback channel of the teleoperated robotic grasping system (Fig. 3). Instead of relying on fixed cameras with limited fields of view, a stereo camera capable of capturing the three-dimensional view of the surroundings was mounted on a pan-tilt platform for simulating human binocular vision. The three-dimensional images captured by the stereo camera were transferred to a VR headset worn by the human operator. The pan-tilt platform can rotate the

stereo camera horizontally and vertically based on the movement of the operator's head using an inertial measurement unit (IMU) fixed on the VR headset, thereby reducing the need to mentally map the environment based on fixed camera views.

Haptic perception for multi-modal feedback channel A teleoperated robotic grasping system was designed to transport and grasp fragile cylindrical fuel products used in current rockets and missiles. This type of product can be easily damaged if the force exerted by the robot's end-effector is not effectively maintained because this type of product is very combustible. The force control strategy implemented in the command channel, which controls the joints of the robot to maintain the desired force level by comparing the desired and exerted forces, should be carefully considered to avoid damage to the fragile product. Traditionally, a data glove is used to capture physical data such as finger bending, and an IMU is attached to capture the global position of the glove (Fig. 4a). Then, these movements are interpreted as control signals that can be executed by the robot. However, most current studies on robot teleoperation do not consider the haptic perception of the feedback channel.

Haptic feedback reduces the cognitive load on the human operator by providing sensory information directly through touch, which is a natural way of interacting with the physical world, thereby leading to a more intuitive teleoperation and allowing the operator to rely on their sense of touch. Haptic glove technology, which was developed in the realm of VR, offers operators a more realistic tactile experience in virtual environments. In this study, in addition to tracking hand movements using bend sensors for the command channel, haptic perception is integrated into the multi-modal feedback channel by borrowing the technology of the haptic glove in the domain of VR (Fig. 4b). On the one hand, this haptic perception allows the human operator to have finer control over the movements of the teleoperated robot, thereby enabling more precise manipulation of objects, especially in the grasping task for fragile products. However, this haptic glove can simulate forces experienced during collisions or contact with objects in a manufacturing environment. This force reflection helps human operators prevent collisions and provides an additional safety mechanism for teleoperation. In the multi-modal feedback channel, a piezoelectric patch attached to the robot's end-effector (i.e., grasper) is used to capture the force exerted on the product. A haptic glove featuring an exoskeleton structure can provide haptic feedback to the fingers of the operator. This haptic glove is equipped with actuators (five small brushless motors controlled using a field-oriented control (FOC) method) that are strategically positioned to exert the same forces captured by the piezoelectric patch by pulling against the fingertips.

Fig. 3 Design of visual perception for multi-modal feedback channel**Fig. 4** Design of haptic perception for multi-modal feedback channel

Auditory perception for multi-modal feedback channel

Auditory perception adds to the naturalness of the teleoperation process because humans are accustomed to perceiving their surroundings through a combination of visual, haptic, and auditory senses in daily life. Although visual and haptic feedbacks are the primary modes of communication for teleoperated robotic grasping systems, the feedback channel more closely resembles how humans interact with the physical world when auditory perception is included. Auditory perception complements these two modalities and reinforces the understanding of the human operator about the remote environment, thereby reducing the cognitive load of the operator and making teleoperation more robust and effective.

Various sounds exist in the manufacturing environment, such as the noises from the robot and its surroundings, and the voices generated by the robotic system that indicate the robot's status, readiness, or completion of specific tasks. Therefore, auditory cues should be selected carefully to avoid using too many auditory cues that can distract the operator, thereby resulting in a high extraneous cognitive load. Inspired

by parking sensors for road vehicles designed to alert drivers of obstacles at blind spots while parking, an alert sound that reminds the human operator of potential collisions between robots and fixed objects in the manufacturing environment will be generated.

The axis-aligned bounding boxes (AABBs) technique was adopted to detect potential collisions between the robot and fixed objects in a manufacturing environment (Wang et al., 2021). AABBs are simple 3D boxes that can be aligned easily with the X-, Y-, and Z-axes of a coordinate system. Potential collisions can be detected by checking if the AABBs of the robot and objects overlap by creating AABBs for robots and fixed objects in a manufacturing environment. A collision will be detected if there is any overlap, and the operators will be notified.

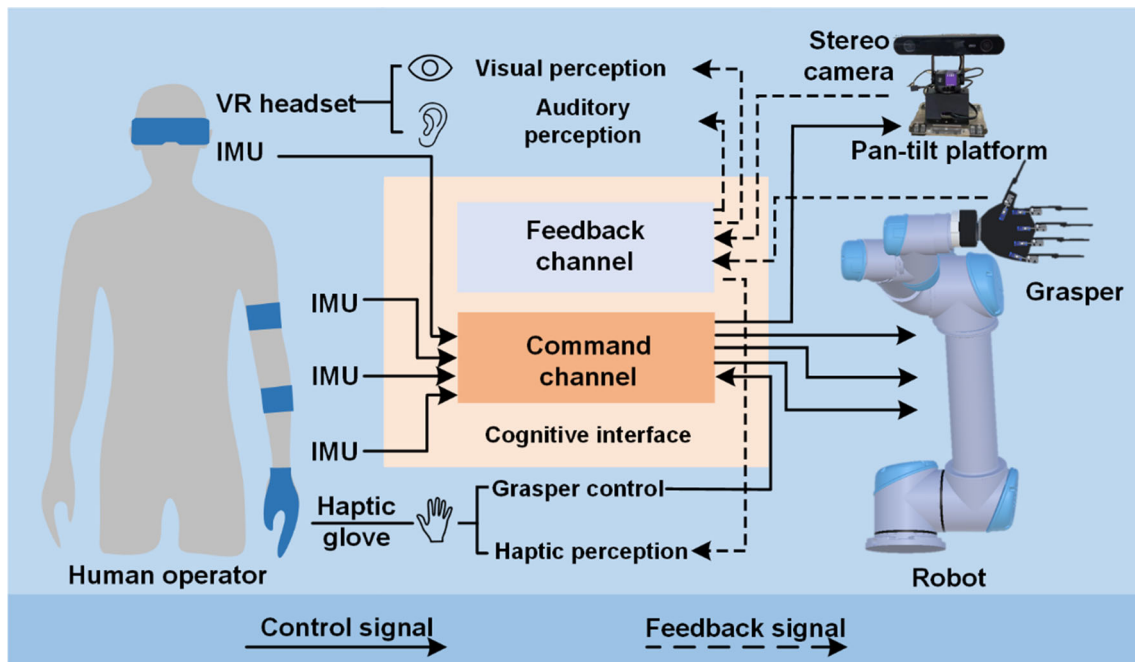


Fig. 5 Architecture of teleoperated robotic grasping system

Implementation of feedback channel in teleoperated robotic grasping system

Architecture of teleoperated robotic grasping system

Figure 5 illustrates the architecture of a teleoperated robotic grasping system. A multi-modal command channel is designed to control the robot's movements and the pan-tilt platform. Two IMUs were fixed on the upper and lower arms to detect the position of the arm and map this position on the robot. A haptic glove was used to capture the bending of fingers, and another IMU attached to the glove was used to capture its position. These movements are interpreted as executable signals that control the actions of the end effector (i.e., the grasper). A multi-modal feedback channel that integrates the visual, haptic, and auditory senses is developed to reduce cognitive load of the human operator.

Visual feedback implementation

A stereo camera mounted on a pan-tilt platform is used to provide dynamic visual feedback by simulating the natural movement of the head of a human operator. A ZED camera, which possesses a vision angle of 110° horizontally and 70° vertically, is selected as the stereo camera in the required teleoperated robotic grasping system because the vision angle of the human operator is about 90° in the horizontal direction and 60° in the vertical direction. The ZED camera provides the human operator with full vision of the manufacturing

environment when teleoperating the actions of the robotic grasping system.

An HTC VIVE Pro, which incorporates an IMU-based dynamic perspective functionality that allows tracking the operator's head movement and provides the necessary data for updating the visual display, was adopted as the VR headset to ensure the alignment between the virtual objects and the perspective of the operator in real time. The dynamic perspective functionality is adapted to follow the movement of the operator's head and control the pan-tilt platform's rotation of the pitch and yaw in this teleoperated robotic grasping system (Fig. 6). The platform implemented in the VR headset is developed based on Unity 3D and Microsoft Visual Studio, which obtains rotations angle increments of the pitch and yaw of the human operator ($\Delta\theta_Y$, $\Delta\theta_Z$) by integrating time with the angular velocity provided by the IMU, and they continuously send them to the two steering gears of the pan-tilt platform to control their angle increments of rotation ($\Delta\theta_{Pitch}$, $\Delta\theta_{Yaw}$).

A closed-loop control system was implemented to adjust the angles of the pan-tilt platform by considering the desired velocity and acceleration limits and maintaining a smooth and controlled motion of the pan-tilt platform to ensure real-time synchronization between the operator's perspective and the stereo camera. The PID control process of the pan-tilt platform for controlling the yaw axis steering gear is shown in Fig. 7. The target angle (θ_{Pitch_new}) is obtained by superimposing the angle increment (θ_Δ) onto the current angle (θ_{Pitch_before}). During the steering gear controlling process, the target angle is converted into the required speed (ω_{Pitch})

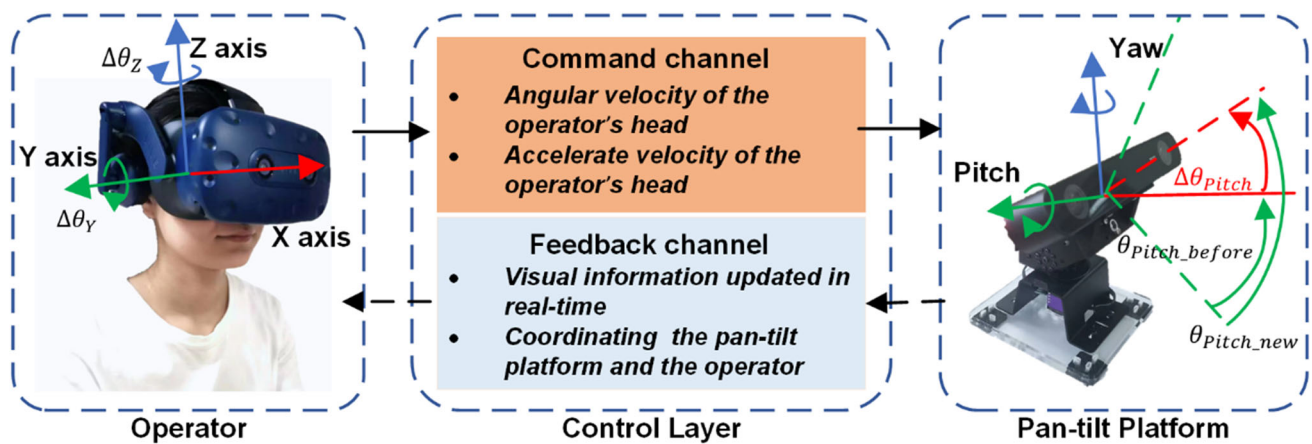


Fig. 6 Process of visual feedback

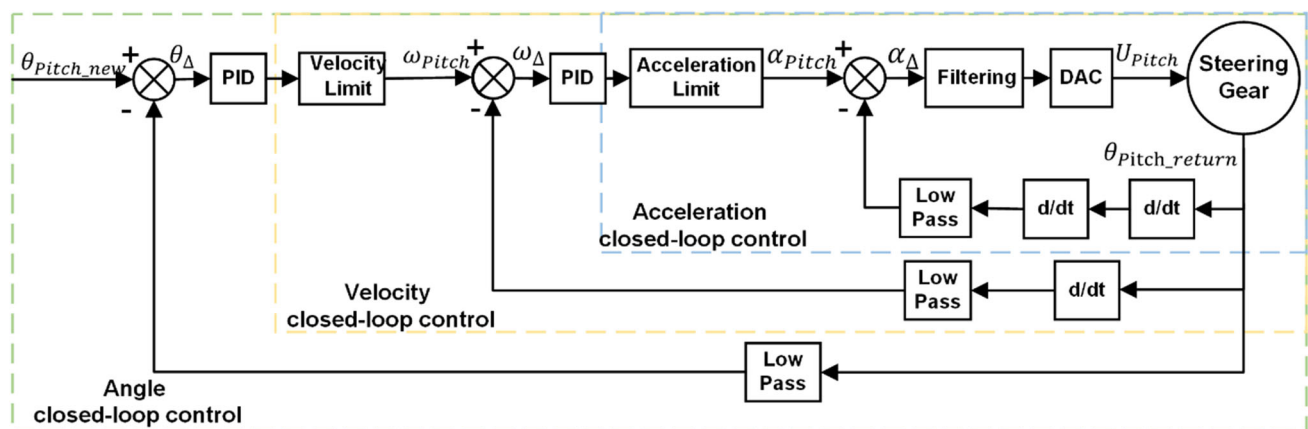


Fig. 7 Steering gear closed-loop control

and acceleration (α_{Pitch}), which can be further converted into the analog voltage signal (U_{Pitch}) for controlling the steering gear. The real-time angle of the steering gear (θ_{Pitch_return}) can be obtained through the position sensor and the average angular velocity. Additionally, the angular acceleration can be determined by deriving the change in angle over time.

Haptic feedback implementation

Haptic feedback plays an important role in reducing the cognitive load of a human operator when conducting a teleoperated grasping task for a fragile product. The haptic glove was initially designed to control the movement of the grasper through a command channel by transmitting the angle information of each finger to the grasper. Inspired by the technology of haptic gloves in the domain of VR, a feedback channel has been developed that enables the human operator to feel the same force exerted on the product, which grants the human operator more precise control over the movements of the teleoperated robot when manipulating fragile products. Besides the robot, the piezoelectric patch stuck on the

grasper of the robot is used for capturing the force exerted on the product, whereas on the side of the human operator, a haptic glove equipped with five brushless motors, featuring an exoskeleton structure, is designed to provide haptic feedback on the fingers of the operator (Fig. 8).

Figure 9 shows the FOC method, which includes two control loops (the angle and current) was implemented in the control unit for achieving precise control of the torques for the five brushless motors. Based on the FOC method, the control currents (I_a , I_b , and I_c) of brushless motors should be decoupled into two orthogonal components: a torque-producing component (q component) that corresponds to the torque-producing current (I_q) and a magnetizing component (d component) that corresponds to the magnetizing current (I_d). The Clark transform converts the three-phase currents (I_a , I_b , and I_c) into two-phase currents (I_α and I_β) aligned with the rotor's magnetic field to achieve this decoupage, and then, the Park transform further rotates the currents I_α and I_β to the reference frame (d axis and q axis) aligned with the position of the rotor. Currents I_q and I_d should be converted into voltages U_q and U_d because brushless motors

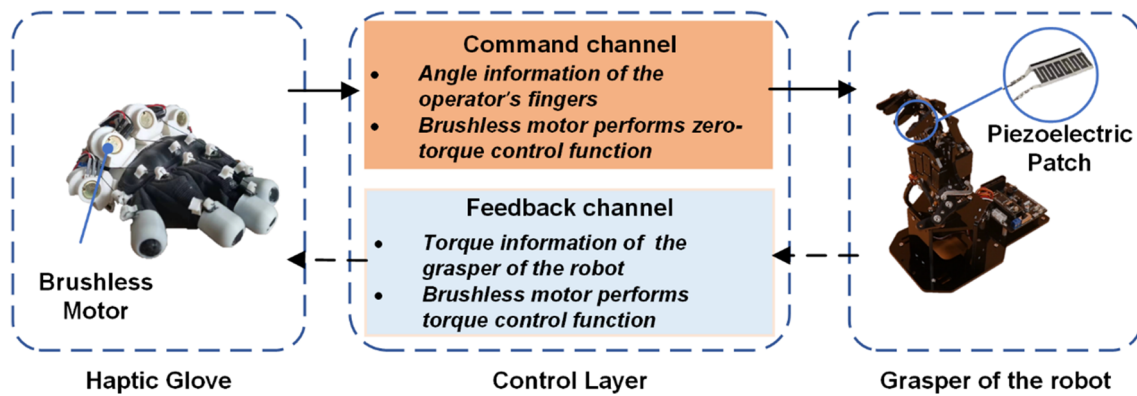


Fig. 8 Process of haptic feedback

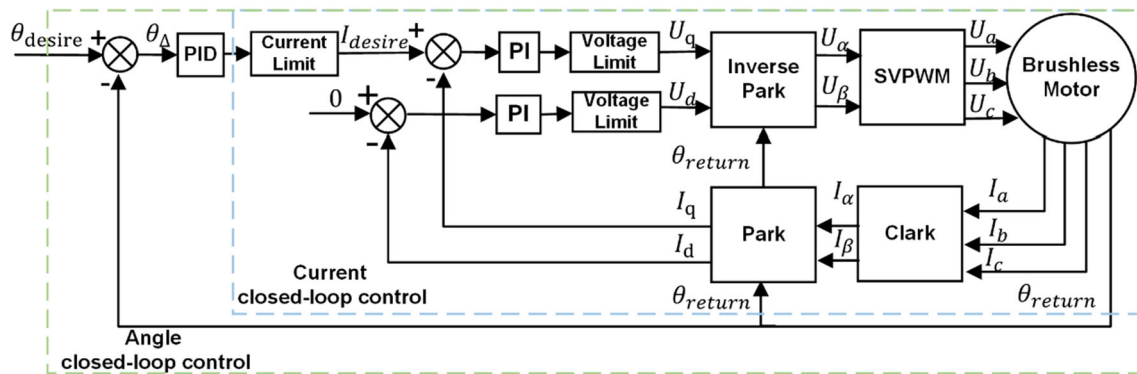


Fig. 9 Brushless motor closed-loop control

are controlled by voltage signals. The inverse Park transform is adopted to transfer the voltages in the rotating reference frame (U_q and U_d) into the voltages in the stationary coordinate system (U_α and U_β). Finally, the two-phase control voltages (U_α and U_β) are converted into the three-phase control voltages (U_a , U_b , and U_c) that effectively control the rotation of the brushless motors using space vector pulse width modulation (SVPWM).

Based on the aforementioned FOC method implemented in the control unit, haptic feedback can be performed using the force detected by the piezoelectric patches fixed on the fingertips of the grasper. Once the piezoelectric patches detect a contact force lower than the predefined threshold, which indicates an insufficient contact force or no contact with the product, the controlling unit limits the currents of the brushless motors to achieve zero-torque control. Although the piezoelectric patches detect a force above the threshold, the system adjusts the currents of the brushless motors to control the torque accordingly.

Auditory feedback implementation

Auditory feedback complements the visual and haptic modalities by reminding the human operator of potential collisions between robots and fixed objects at blind spots, thereby

making teleoperation more robust and effective. In the teleoperated robotic grasping system, the technique of axis-Aligned bounding boxes (AABBs) was adopted to detect potential collisions between the robot and fixed objects (i.e., protective fences).

A universal robot (UR5), a six-degree-of-freedom collaborative robot, is used in the teleoperated robotic grasping system required to move the grasper. Consequently, seven AABBs are created for the seven links (including the grasper) of the robotic grasping system. The created AABBs should be updated continuously to match the robotic grasping system because the robot and grasper move and change their positions and orientations during teleoperation. However, this requires the recalculation of the minimum and maximum extents of each AABB based on the geometric and kinematic parameters of the robot and grasper. The Denavit–Hartenberg (D–H) modeling approach, which is a systematic method to define the reference frame of a robot, is used to describe the geometric and kinematic parameters of a robot's joints and links.

Equation (1) presents the transformation matrix used to describe the relationship between consecutive reference frames (from frame $\{i-1\}$ to frame $\{i\}$) attached to the robot links.

$${}_{i-1}^i T = \begin{bmatrix} \cos \theta_i - \sin \theta_i \cos \alpha_i \sin \theta \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & b_i \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad (1)$$

where a , α , b and θ denote the link length, link twist, link offset, and joint angle of the two consecutive frames, respectively. Once transformation matrices between frames are determined, the position and orientation of each link of the robot and end-effector (i.e., grasper) are calculated iteratively with respect to the base frame by applying Eq. (1). The transformation matrix between the robotic grasping system of the grasper frame and robot base frame can be calculated as

$${}_{0}^{grasper} T = {}_{0}^{grasper} T_5^6 T_4^5 T_3^4 T_2^3 T_1^2 T_0^1 T, \quad (2)$$

where ${}_{i-1}^i T$ represents the transformation matrix from the coordinate system of the frame $\{i-1\}$ to frame $\{i\}$.

The smallest and largest coordinates along the three axes (X , Y , and Z) that encapsulate the entire volume of each part of the robotic grasping system (i.e., the six links and grasper) at any moment can then be calculated using the minimum and maximum extents of the geometry of each part for each axis. The AABB for each part can be obtained using the minimum and maximum extents obtained in the previous step. AABB is defined by two points: the minimum corner (X_{Min} , Y_{Min} , and Z_{Min}) and the maximum corner (X_{Max} , Y_{Max} , and Z_{Max}), where the former represents the point with the smallest coordinates along all three axes, while the latter represents the point with the largest coordinates along all three axes. The AABBs of the object (e.g., the protective fences in this teleoperated grasping scenario) in the manufacturing environment can be constructed following the same steps (Fig. 10a).

An algorithm based on the separating axis theorem (SAT) is used to check the collisions between the AABBs of the moving parts of the robotic grasping system and those of other objects in the environment. The SAT algorithm checked the distance along the selected axes when UR5 moved according to the instructions of the human operator. In the three-dimensional space, 15 separating axes were selected, which include six axes along the directions of the facets of the two AABBs and nine along the directions of the cross products of the vectors of any two facets of the AABBs. When the distance between the AABBs of any movement parts of the robotic grasping system and the AABBs of the protective fences do not overlap and are below the safety threshold, the VR headset offers a warning tone to the human operator as reminder for adjusting the robot pose in time (Fig. 10b). Further, if there is any overlap, the teleoperated robotic grasping system stops the motion of UR5 and alerts the human operators of a collision between the AABBs of the robot and the

AABBs of the protective fences, thereby avoiding collisions in remote natural environments.

Experiment and evaluation

Experimental setup

A series of experiments are performed to validate the proposed design approach for the multi-modal feedback channel in reducing the human operators' cognitive load. Besides, this paper evaluates the cognitive load by merging the objective aspects of experimental results with the subjective aspects of operator feedback.

A total of six groups of experiments are designed using different feedback channels to conduct teleoperated grasping tasks. According to SubSection “[Architecture of teleoperated robotic grasping system](#)”, human operators can use multiple pieces of equipment for acquiring different feedback information, including a VR headset for visual and auditory information and a haptic glove for haptic information. During the experiments, the human operators were instructed to remain within a safe workspace separate from the robotic system. Figure 11a illustrates an industrial camera that provides a bird's-eye view as the fixed visual feedback channel. The second experiment is presented in Fig. 11b, wherein the stereo camera was fixed on the pan-tilt platform to provide the human operator with dynamic visual feedback by combining it with the VR headset. Figure 11c shows the experiment where the human operator is provided with haptic feedback based on fixed visual feedback. In the experimental setup shown in Fig. 11d, auditory feedback was introduced in addition to dynamic visual feedback to remind the human operator of the potential collisions between the robots and protective fences at blind spots. Figure 11e shows the feedback channel, which combines haptic feedback to sense the fragile product with dynamic visual feedback; however, the auditory feedback is not integrated. The experimental setup for implementing the proposed multi-modal feedback channel, in which the visual, haptic, and auditory senses are integrated, is illustrated in Fig. 11f.

A total of 24 participants were selected and divided into three groups based on their level of teleoperation experience. Participants in groups A, B, and C had sufficient experience in teleoperation using VR headsets and a data glove, little experience using VR headsets and data gloves, and were novices who had only a preliminary understanding of robot teleoperation, respectively. The participants randomly performed the experiments in twenty-four different orders to eliminate bias caused by the order of the six types of experimental setups. Further, the participants had to conduct the same teleoperated grasping task using the six experiments mentioned above.

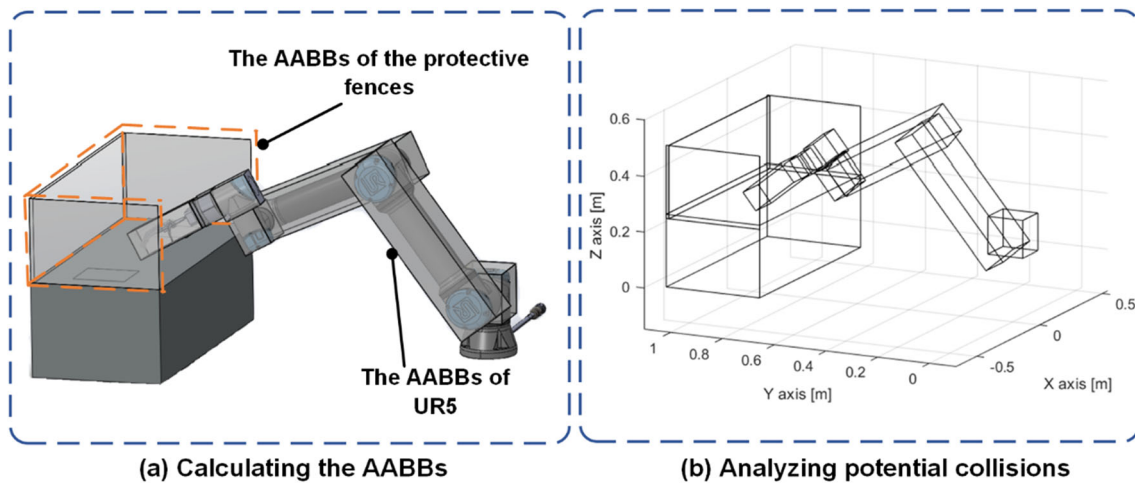


Fig. 10 Process of detecting potential collisions

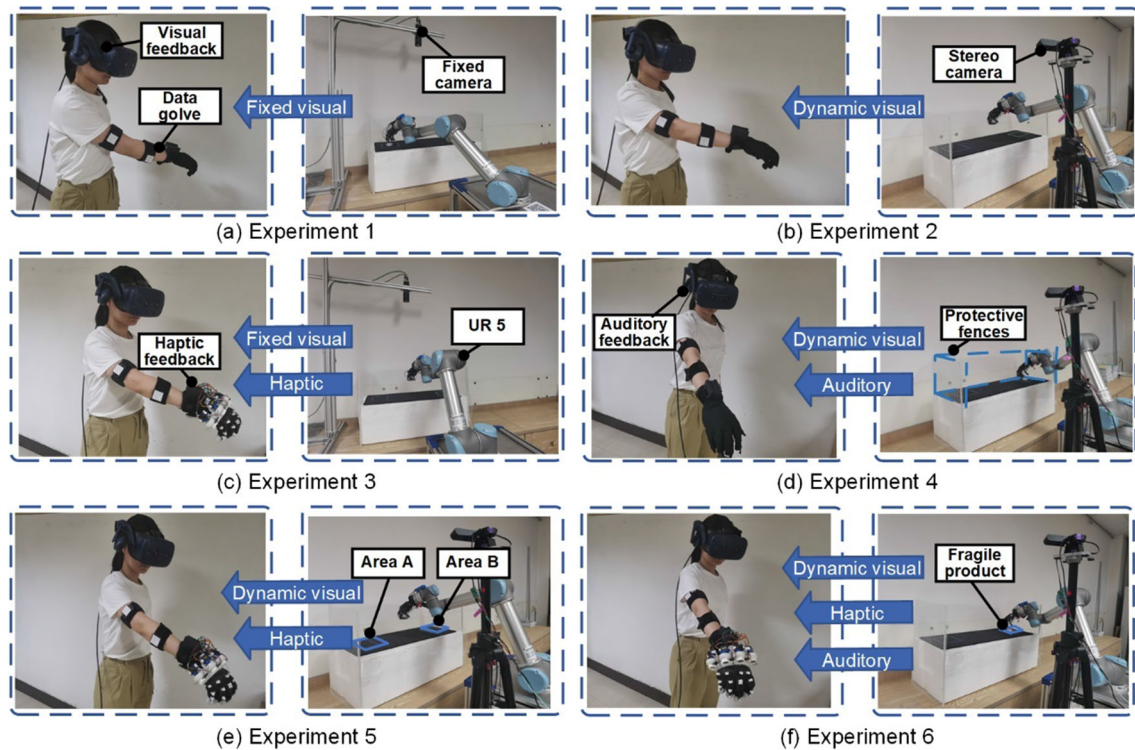


Fig. 11 Six groups of experiments using different feedback channels

The teleoperated grasping task is divided into three steps: (1) The operator remotely guides the robot towards area A and then instructs the robot to grasp the fragile product and maintain a certain height. (2) The operator maintains the stable grasper of the robot and moves the product from area A to area B. (3) When the robot reaches area B, the operator controls the robot to place the fragile product.

Each operator was given ten opportunities to complete the grasping task based on six types of experimental setups. The operator was asked to rest for 10 min between experiments

to avoid mental fatigue caused by the teleoperation task. The objective and subjective evaluation approaches were used to compare the proposed multi-modal feedback channel with other feedback channels.

Objective evaluation of cognitive load for different feedback channels

The objective evaluation of cognitive load refers to the quantitative measurement and evaluation of the cognitive load

experienced by participants. Two indicators, task completion time (TCT) and task completion rate (TCR), are widely used for analyzing the cognitive load of participants (Muke et al., 2022). Therefore, TCT and TCR values were recorded in different experiments. In addition, TCT and TCR values do not consider failed cases because they consider only successful experimental results. This study further analyzes the reasons for the failed experimental cases.

TCT

Figure 12 illustrates the TCT data obtained by comparing the overall distributions of the six experiments. Table 3 presents the mean and standard deviation of the TCT, where the mean (M) indicates the central tendency of the TCT during the experiment, and the standard deviation (SD) assesses the spread tendency of the TCT when operators complete the different experiments. The experimental results showed that Experiment 1 (Exp 1) had the longest TCT. The comparison results between Experiments 2 (Exp 2) and 3 (Exp 3) indicate that the integration of dynamic visual feedback is more beneficial for reducing the TCT. In addition to dynamic visual feedback, Experiment 4 (Exp 4), which incorporated auditory feedback, achieved a more dispersed distribution of TCT than that in Experiment 5 (Exp 5), wherein haptic feedback was integrated. Compared to the first five experiments, Experiment 6 (Exp 6) showed the shortest average TCT; however, the fluctuation in the time distribution in Experiment 6 (Exp 6) was more significant than that in Experiment 5 (Exp 5).

TCR

Table 4 presents the comparison results of TCR for the different groups and experiments. The first three experiments reveal that the TCR improved when both dynamic visual and haptic feedback were considered. The haptic feedback is more effective in enhancing the TCR than dynamic visual feedback. Further, Exp 6, in which the proposed multi-modal feedback channel is implemented exhibit the highest TCR compared to the other five experiments. The comparison results among the different rows indicate that participants from groups B and C in Exp 6 demonstrate a rapid adaptation to accomplish the required teleoperation task.

Failure reasons

An analysis is conducted to identify the causes of failure in each experiment, which can be categorized into grasping failure, which is the failure to grasp the product during the first step of the grasping task; product drop, wherein the product falls from the robot's grasper during the grasping task; product deformation, where significant deformation is detected in the product after completing the grasping task; and robot

collision, where the AABBs of the robot or grasper collides with the AABBs of the protective fences during the grasping task. Table 5 lists the number of failures in all experiments.

The results of analyzing the reasons for failure between Exp 1 and the other five experiments reveal that dynamic visual feedback plays an important role in avoiding grasping failure, haptic feedback reduces the number of product deformation, and that auditory feedback decreases the number of robot collisions. Exp 6 effectively reduced the number of all types of experimental failure by implementing the proposed multi-modal feedback channel.

Subjective evaluation of cognitive load for different feedback channels

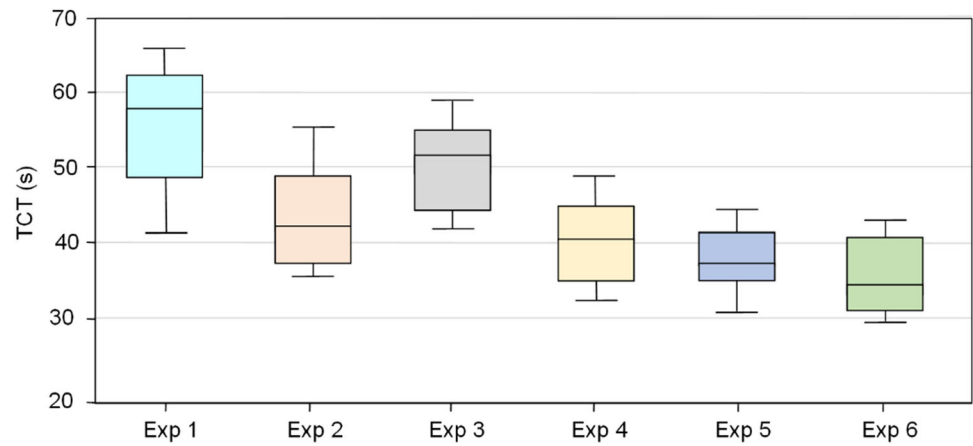
Subjective assessments enable participants to express the subtleties of their cognitive load during teleoperation, which may not be entirely captured by objective evaluations. With the NASA Task Load Index (NASA-TLX) and a subjective rating scale, participants can provide more details and identify hidden factors when assessing their cognitive loads.

NASA-TLX

The NASA-TLX can comprehensively assess the cognitive load experienced by participants (Nguyen & Zeng, 2017). Figure 13 illustrates the overall cognitive load scores of the participants for the experiment, which were determined using the NASA-TLX method. The scores of the participants in Group A indicated that they experienced the lowest overall cognitive load in Exp 6. Compared to the participants in Group C, those in Group B demonstrated a lower overall cognitive load. Table 6 illustrates the M and SD for evaluating the central and spread tendencies of overall cognitive load. The score of Exp 6 exhibited a significant reduction compared to that of the other experiments.

Subjective rating scale

A subjective rating scale is a common approach employed to measure the cognitive load. Thees et al. (2021) proposed a 7-point Likert scale that can be considered a subjective rating scale to rate cognitive loads. A new subjective rating scale corresponding to the experiment was proposed inspired by the 7-point Likert scale. This new scale comprises two questions addressing intrinsic cognitive load, three focusing on extraneous cognitive load, and three on germane cognitive load (Online Appendix A). All items were rated from 1 (completely wrong) to 7 (absolutely right), and the scores for the three cognitive loads are presented in Fig. 14. The results showed that Exp 6 had significantly lower intrinsic and extrinsic cognitive loads than the other experiments, and

Fig. 12 Box and whisker plot of TCT**Table 3** Mean and standard deviation of TCT

	Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	Exp 6
M	56.25	42.65	50.42	40.29	37.75	36.54
SD	7.17	6.05	5.24	5.19	4.16	4.56

Table 4 Comparison results of TCR for different groups and experiments

	Exp 1 (%)	Exp 2 (%)	Exp 3 (%)	Exp 4 (%)	Exp 5 (%)	Exp 6 (%)
Group A	88.75	91.25	93.75	95	97.5	98.75
Group B	85	86.25	88.75	92.5	95	96.25
Group C	80	83.75	85	90	91.25	93.75

Table 5 Failure reasons of different experiments

	Exp 1	Exp 2	Exp3	Exp4	Exp 5	Exp 6
Grasping Failure	9	5	7	4	2	2
Product Drop	10	8	4	5	3	2
Product Deformation	8	9	4	6	3	3
Robot Collision	10	9	11	3	5	2

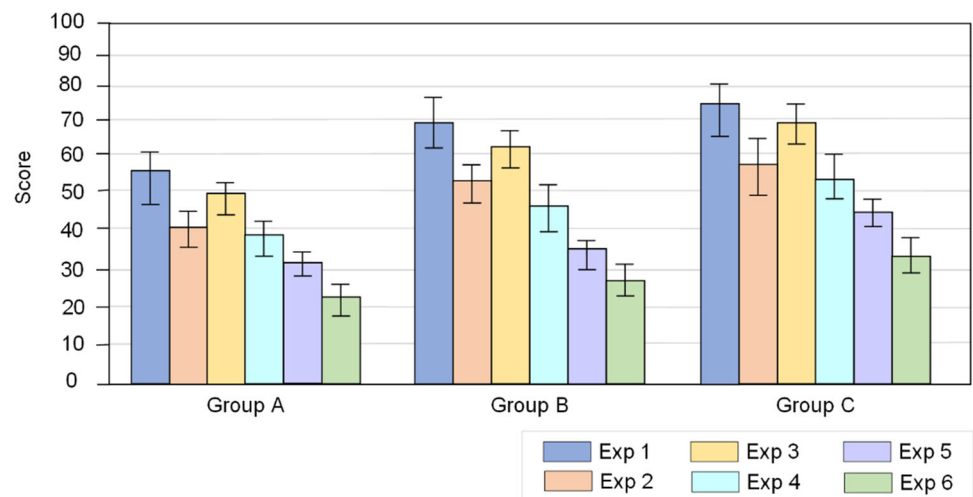
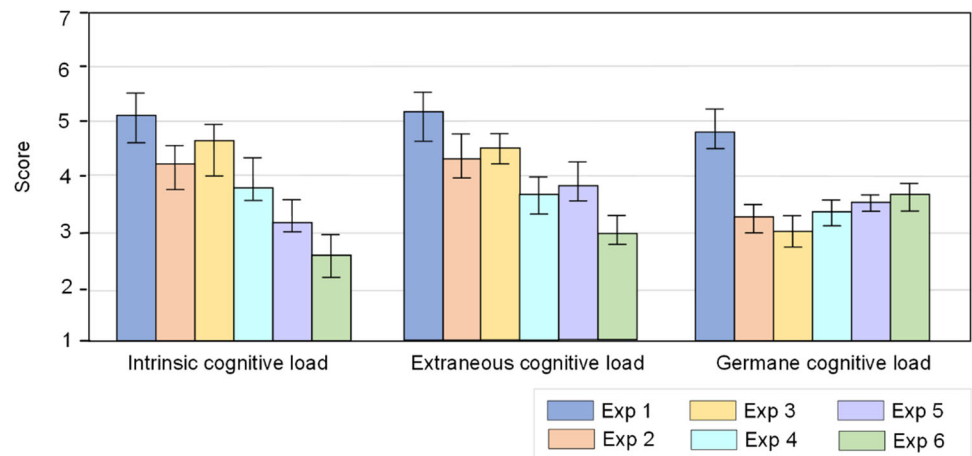
Fig. 13 Scores of overall cognitive load

Table 6 Mean and standard deviation of overall cognitive load

	Exp 1	Exp 2	Exp 3	Exp 4	Exp 5	Exp 6
M	68.38	51.71	59.25	47.58	35.54	31.46
SD	9.19	7.87	8.62	7.27	5.55	5.69

Fig. 14 Scores of three cognitive loads

the germane cognitive load showed a relatively unchanged or even increasing trend compared to the other experiments.

Interviews of participants

Conducting interviews is a valuable method to gain further insight into the cognitive load of the participants during the experiments. Several open-ended questions (Online Appendix B) were used to encourage participants to provide specific events in which they perceived a certain cognitive load or any additional comments or suggestions regarding the experiments.

- (1) *Intrinsic cognitive load* Group C participants unanimously supported the opinion that they needed to mentally map the fixed visual perspective to their own spatial awareness and adjust the actions of the robot based on the restricted viewpoint because they had to constantly monitor the status of fragile products. Seventeen participants reported that significant mental demands were imposed when engaging in the grasping task without haptic feedback. The majority of the participants (19) strongly perceived that the proposed multi-modal feedback channel could provide intuitive and direct information about the manufacturing environment.
- (2) *Extraneous cognitive load* Most participants (22) considered that additional effort was required to comprehend the coordination between the visual and haptic information when conducting experiments utilizing

both fixed visual and haptic feedback. Twenty participants believed that dynamic visual feedback could provide them with natural environmental information and lower their extraneous cognitive load. In addition, the participants in Group A commented on the proposed auditory feedback channel and stated that the alert sound delivered only the necessary auditory cues without imposing significant cognitive burden.

- (3) *Germane cognitive load* During the initial five experiments, Group A participants reported that they quickly completed the grasping tasks based on their previous knowledge. Group B participants indicated that it was easy to complete the experiment with support from the proposed multi-modal feedback channel; however, the participants in Group C indicated that they needed more practice before they could complete the experiment proficiently.

Discussion

This study focuses on the design of feedback channels in a cognitive interface in which multi-modal sensory data are integrated to reduce the cognitive load of human operators during the teleoperated robotic manufacturing process. The advantages of the proposed multi-modal feedback channel are summarized as follows:

Reduction of the intrinsic cognitive load

The level of intrinsic cognitive load throughout the teleoperated robotic manufacturing process relies on the complexity inherent in the manufacturing task. Consequently, a human operator can face a substantial intrinsic cognitive load while attempting to comprehend and anticipate the behavior of the robot in an intricate and ever-changing environment. This study adopts the mapping principle to establish a clear correspondence between the robot and the human operators so that the intrinsic cognitive load on human operators can be minimized.

Reduction of the extraneous cognitive load

Extraneous cognitive load during the teleoperated robotic manufacturing process is imposed by irrelevant or unnecessary information that distracts the manufacturing task. This cognitive load has been reduced by introducing several improvements to make the grasping process more natural, intuitive, and efficient. For example, visual feedback is provided for an immersive and adaptive perspective that replicates the natural movement of a human operator's head in alignment with the perceptual and behavioral patterns. Further, haptic feedback is employed to reduce the extraneous cognitive load on human operators by providing sensory information directly through touch. Finally, auditory feedback is employed to raise an alarm to alert human operators of remote environment information, thereby avoiding the effect of actual field noise.

Increase in the germane cognitive load

Germane cognitive load is beneficial because it supports long-term memory and increases during the teleoperation process of manufacturing robotic systems. Constructing a new mental model that can be understood easily by operators and integrated into the teleoperation process to form a coherent understanding of the teleoperation task can contribute to a germane cognitive load. A mental model was constructed based on natural information that human operators acquire from various senses during interactions with a teleoperated robot. Visual cues provide a sense of spatial layout, haptic feedback imparts tactile information, and auditory cues offer additional contextual details regarding sounds and events. Together, these cues provide a more complete picture, thereby leading to a more natural and immersive experience for the operator, which helps them make better-informed decisions.

Based on the advantages of the proposed multi-modal feedback channel, this study approaches the teleoperation process from a human perspective to reduce the cognitive load on humans during manufacturing tasks. The objective

experimental data indicate that the multi-modal feedback channel improves the overall performance in the teleoperation process. The multi-modal feedback channel proposed in this study has substantial potential for application in hazardous and unstructured manufacturing environments, especially in precise and concentrated domains. However, unpredictable situations and limitations may still exist in teleoperated robotic manufacturing processes because of the complexity of human cognitive processes and the uncertainty of industrial manufacturing environments. Consequently, the proposed approach must be expanded from the following perspectives:

Individual differences

The experimental results indicate that individual differences, such as cognitive abilities, perceptual preferences, and prior experience, can affect the influence of multi-modal information on cognitive processes. Individuals can have different preferences and capabilities in processing information from different modalities. If the information aligns well with the cognitive processes and mental models of the operator, it is more likely to enhance cognitive processing. However, if information is presented in a manner that conflicts with the cognitive processes or mental representations of the operator, it can hinder cognitive performance. Therefore, what works well for one person may not work effectively for another. These individual differences should be considered in future studies.

Conflicts between multi-modal information

Integrating various modalities can result in a more comprehensive understanding of tasks or scenarios when they provide complementary information. However, sometimes, the information provided through different modalities may conflict with each other, increasing the cognitive load rather than enhancing the cognitive processes. A conflict-detection and warning system must be developed and implemented in teleoperated robotic manufacturing systems to alert operators when conflicting information is detected to solve this problem. These warnings can prompt the operator to reassess the situation and initiate appropriate action. However, human operators need to be provided with adequate training and familiarization sessions to help them better understand how different sensory cues correspond to each other and how to interpret and reconcile conflicting information. When multi-modal information conflicts occur, it can help identify the most critical sensory cues for the specific task and prioritize their presentation based on their training experience.

Conclusion

This study presented a design method for the feedback channel of a teleoperated robotic grasping system in terms of human cognition, which aims to reduce the cognitive load on human operators during teleoperation in intricate manufacturing environments. Based on CLT, different cognitive loads during the teleoperated robotic manufacturing process were clarified and analyzed, resulting in improvements in the design of the feedback channel. In accordance with the design improvement steps, the mapping principle was first adopted to select sensory data with multiple modalities by aligning the sensory data of the robot's actions and current state of the workspace with the cognitive abilities of the human operator. Subsequently, by utilizing the teleoperated robotic grasping system as a case study, capturing approaches with their implementation details for the three types of selected sensory data were presented. A series of verification tests involving 24 participants were conducted to quantitatively and qualitatively evaluate the new feedback channel using grasping tasks for fragile products. The results indicate the effectiveness of the new feedback channel in improving manufacturing efficiency and reducing the cognitive load of human operators.

Future research can be divided into two parts: (1) Individual difference should be studied more extensively. Continuously testing the design of the feedback channel with a real operator and gather feedback for iterative improvements is necessary even though this study improved the design of the feedback channel in a teleoperated robotic grasping system from the perspective of human cognition and attempted to make the teleoperation process more natural, intuitive, and efficient, considering the cognitive abilities, perceptual preferences, and prior experience of different operators. (2) A conflict-detection and warning system must be developed to solve the conflict problem of multi-modal information captured by different sensors. Adequate training and familiarization sessions should be provided to human operators.

Supplementary Information The online version contains supplementary material available at <https://doi.org/10.1007/s10845-024-02451-x>.

Acknowledgements This work was supported by the Natural Science Basic Research Program of Shaanxi (Grant No. 2023-JC-YB-299), China Postdoctoral Science Foundation (Grant No. 2022M712933) and National Natural Science Foundation of China (Grant No. 51805437).

Author contributions Conceptualization: Zhanxi Wang; Methodology: Chen Zheng; Formal analysis: Shiqi Gao; Investigation: Shiqi Gao, Yang Yu; Writing—original draft preparation: Chen Zheng, Kangning Wang; Software: Kangning Wang; Data curation: Yang Yu; Writing—review & editing: Yunlong Tang; Resources: Zhanxi Wang; Supervision: Yunlong Tang.

Funding Open Access funding enabled and organized by CAUL and its Member Institutions.

Data availability No data was used for the research described in the article.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by/4.0/>.

References

- Almeida, L., Menezes, P., & Dias, J. (2020). Interface transparency issues in teleoperation. *Applied Sciences*, 10(18), 6232. <https://doi.org/10.3390/app10186232>
- Anggraeni, D., Sanjaya, W. S. M., Nurasyidiek, M. Y. S., & Munawwaroh, M. (2018). The implementation of speech recognition using mel-frequency cepstrum coefficients (MFCC) and support vector machine (SVM) method based on Python to control robot arm. *IOP Conference Series: Materials Science and Engineering*, 288, 012042. <https://doi.org/10.1088/1757-899X/288/1/012042>
- Arents, J., & Greitans, M. (2022). Smart industrial robot control trends, challenges and opportunities within manufacturing. *Applied Sciences*, 12(2), 937. <https://doi.org/10.3390/app12020937>
- Baker, J., Jones, D., & Burkman, J. (2009). Using visual representations of data to enhance sensemaking in data exploration tasks. *Journal of the Association for Information Systems*, 10(7), 533–559. <https://doi.org/10.17705/1jais.00204>
- Bandala, M., West, C., Monk, S., Montazeri, A., & Taylor, C. J. (2019). Vision-based assisted tele-operation of a dual-arm hydraulically actuated robot for pipe cutting and grasping in nuclear environments. *Robotics*, 8(2), 42. <https://doi.org/10.3390/robotics8020042>
- Bejczy, B., Bozyil, R., Vaičekauskas, E., Krogh Petersen, S. B., Bøgh, S., Hjorth, S. S., & Hansen, E. B. (2020). Mixed reality interface for improving mobile manipulator teleoperation in contamination critical applications. *Procedia Manufacturing*, 51, 620–626. <https://doi.org/10.1016/j.promfg.2020.10.087>
- Brassetti, A., Ragusa, A., Tedesco, F., Prata, F., Cacciatore, L., Iannuzzi, A., et al. (2023). Robotic surgery in urology: History from PROBOT® to HUGOTM. *Sensors*, 23(16), 7104. <https://doi.org/10.3390/s23167104>
- Castro, A., Silva, F., & Santos, V. (2021). Trends of human–robot collaboration in industry contexts: handover, learning, and metrics. *Sensors*, 21(12), 4113. <https://doi.org/10.3390/s21124113>
- Chacón, A., Ponsa, P., & Angulo, C. (2021). Cognitive interaction analysis in human–robot collaboration using an assembly task. *Electronics*, 10(11), 1317. <https://doi.org/10.3390/electronics10111317>

- Chamorro, S., Collier, J., & Grondin, F. (2021). Neural Network Based Lidar Gesture Recognition for Realtime Robot Teleoperation. In *2021 IEEE International Symposium on Safety, Security, and Rescue Robotics (SSRR)* (pp. 98–103). IEEE. <https://doi.org/10.1109/SSRR53300.2021.9597855>
- Darvish, K., Penco, L., Ramos, J., Cisneros, R., Pratt, J., Yoshida, E., et al. (2023). Teleoperation of Humanoid Robots: A Survey. *IEEE Transactions on Robotics*, 39(3), 1706–1727. <https://doi.org/10.1109/TRO.2023.3236952>
- Ding, R., Cheng, M., Han, Z., Wang, F., & Xu, B. (2022). Human-machine interface for a master-slave hydraulic manipulator with vision enhancement and auditory feedback. *Automation in Construction*, 136, 104145. <https://doi.org/10.1016/j.autcon.2022.104145>
- Du, G., Zhang, P., Mai, J., & Li, Z. (2012). Markerless kinect-based hand tracking for robot teleoperation. *International Journal of Advanced Robotic Systems*, 9(2), 36. <https://doi.org/10.5772/50093>
- Ehsan, U., Harrison, B., Chan, L., & Riedl, M. O. (2018). Rationalization. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society* (pp. 81–87). ACM. <https://doi.org/10.1145/3278721.3278736>
- Fang, B., Wei, X., Sun, F., Huang, H., Yu, Y., & Liu, H. (2019). Skill learning for human–robot interaction using wearable device. *Tsinghua Science and Technology*, 24(6), 654–662. <https://doi.org/10.26599/TST.2018.9010096>
- Gervasi, R., Mastrogiovanni, L., & Franceschini, F. (2020). A conceptual framework to evaluate human–robot collaboration. *The International Journal of Advanced Manufacturing Technology*, 108(3), 841–865. <https://doi.org/10.1007/s00170-020-05363-1>
- Huang, X., Halwani, M., Muthusamy, R., Ayyad, A., Swart, D., Seneviratne, L., et al. (2022). Real-time grasping strategies using event camera. *Journal of Intelligent Manufacturing*, 33(2), 593–615. <https://doi.org/10.1007/s10845-021-01887-9>
- Javaid, M., Haleem, A., Singh, R. P., & Suman, R. (2021). Substantial capabilities of robotics in enhancing industry 4.0 implementation. *Cognitive Robotics*, 1, 58–75. <https://doi.org/10.1016/j.cogr.2021.06.001>
- Jeong, S., Kim, J., & Lee, J. (2024). The differential effects of multisensory attentional cues on task performance in VR depending on the level of cognitive load and cognitive capacity. *IEEE Transactions on Visualization and Computer Graphics*. <https://doi.org/10.1109/TVCG.2024.3372126>
- Ji, W., & Wang, L. (2019). Industrial robotic machining: A review. *The International Journal of Advanced Manufacturing Technology*, 103(1–4), 1239–1255. <https://doi.org/10.1007/s00170-019-03403-z>
- Keshavarparast, A., Battini, D., Battaia, O., & Pirayesh, A. (2023). Collaborative robots in manufacturing and assembly systems: Literature review and future research agenda. *Journal of Intelligent Manufacturing*. <https://doi.org/10.1007/s10845-023-02137-w>
- Koelwijn, T., Shinn-Cunningham, B. G., Zekveld, A. A., & Kramer, S. E. (2014). The pupil response is sensitive to divided attention during speech processing. *Hearing Research*, 312, 114–120. <https://doi.org/10.1016/j.heares.2014.03.010>
- Kolbeinsson, A., Lagerstedt, E., & Lindblom, J. (2019). Foundation for a classification of collaboration levels for human–robot cooperation in manufacturing. *Production & Manufacturing Research*, 7(1), 448–471. <https://doi.org/10.1080/21693277.2019.1645628>
- Kong, X. T. R., Luo, H., Huang, G. Q., & Yang, X. (2019). Industrial wearable system: The human-centric empowering technology in Industry 4.0. *Journal of Intelligent Manufacturing*, 30(8), 2853–2869. <https://doi.org/10.1007/s10845-018-1416-9>
- Kou, H., Shi, H., & Zhao, H. (2023). Subspace and second-order statistical distribution alignment for cross-domain recognition of human hand motions. *Journal of Intelligent Manufacturing*. <https://doi.org/10.1007/s10845-023-02150-z>
- Li, S., Zheng, P., Liu, S., Wang, Z., Wang, X. V., Zheng, L., & Wang, L. (2023). Proactive human–robot collaboration: Mutual-cognitive, predictable, and self-organising perspectives. *Robotics and Computer-Integrated Manufacturing*, 81, 102510. <https://doi.org/10.1016/j.rcim.2022.102510>
- Liu, M., Lv, J., Du, S., Deng, Y., Shen, X., & Zhou, Y. (2024). Multi-resource constrained flexible job shop scheduling problem with fixture-pallet combinatorial optimisation. *Computers & Industrial Engineering*, 188, 109903. <https://doi.org/10.1016/j.cie.2024.109903>
- Liu, R., & Zhang, X. (2019). A review of methodologies for natural-language-facilitated human–robot cooperation. *International Journal of Advanced Robotic Systems*, 16(3), 172988141985140. <https://doi.org/10.1177/1729881419851402>
- Mahmud, S., Lin, X., & Kim, J.-H. (2020). Interface for Human Machine Interaction for assistant devices: A Review. In *2020 10th Annual Computing and Communication Workshop and Conference (CCWC)* (pp. 0768–0773). IEEE. <https://doi.org/10.1109/CCWC47524.2020.9031244>
- Moniruzzaman, M., Rassau, A., Chai, D., & Islam, S. M. S. (2022). Teleoperation methods and enhancement techniques for mobile robots: A comprehensive survey. *Robotics and Autonomous Systems*, 150, 103973. <https://doi.org/10.1016/j.robot.2021.103973>
- Muke, P. Z., Telec, Z., & Trawiński, B. (2022). *Multimodal approach to measuring cognitive load using Sternberg memory and input diagrammatic reasoning tests* (pp. 693–713). Springer. https://doi.org/10.1007/978-3-031-21967-2_56
- Mustafa, M. K., Allen, T., & Appiah, K. (2019). A comparative review of dynamic neural networks and hidden Markov model methods for mobile on-device speech recognition. *Neural Computing and Applications*, 31(S2), 891–899. <https://doi.org/10.1007/s00521-017-3028-2>
- Naceri, A., Mazzanti, D., Bimbo, J., Praticchizzo, D., Caldwell, D. G., Mattos, L. S., & Deshpande, N. (2019). Towards a Virtual Reality Interface for Remote Robotic Teleoperation. In *2019 19th International Conference on Advanced Robotics (ICAR)* (pp. 284–289). IEEE. <https://doi.org/10.1109/ICAR46387.2019.8981649>
- Nguyen Ngoc, H., Lasa, G., & Iriarte, I. (2022). Human-centred design in industry 4.0: case study review and opportunities for future research. *Journal of Intelligent Manufacturing*, 33(1), 35–76. <https://doi.org/10.1007/s10845-021-01796-x>
- Nguyen, T. A., & Zeng, Y. (2017). Effects of stress and effort on self-rated reports in experimental study of design activities. *Journal of Intelligent Manufacturing*, 28(7), 1609–1622. <https://doi.org/10.1007/s10845-016-1196-z>
- Pan, Y., Chen, C., Li, D., Zhao, Z., & Hong, J. (2021). Augmented reality-based robot teleoperation system using RGB-D imaging and attitude teaching device. *Robotics and Computer-Integrated Manufacturing*, 71, 102167. <https://doi.org/10.1016/j.rcim.2021.102167>
- Qiu, S., Li, Z., He, W., Zhang, L., Yang, C., & Su, C.-Y. (2017). Brain-machine interface and visual compressive sensing-based teleoperation control of an exoskeleton robot. *IEEE Transactions on Fuzzy Systems*, 25(1), 58–69. <https://doi.org/10.1109/TFUZZ.2016.2566676>
- Sepp, S., Howard, S. J., Tindall-Ford, S., Agostinho, S., & Paas, F. (2019). Cognitive load theory and human movement: Towards an integrated model of working memory. *Educational Psychology Review*, 31(2), 293–317. <https://doi.org/10.1007/s10648-019-09461-9>
- Solanes, J. E., Muñoz, A., Gracia, L., Martí, A., Gírbés-Juan, V., & Tornero, J. (2020). Teleoperation of industrial robot manipulators based on augmented reality. *The International Journal of Advanced Manufacturing Technology*, 111(3–4), 1077–1097. <https://doi.org/10.1007/s00170-020-05997-1>

- Su, H., Qi, W., Chen, J., Yang, C., Sandoval, J., & Laribi, M. A. (2023). Recent advancements in multimodal human–robot interaction. *Frontiers in Neurobotics*, *17*. <https://doi.org/10.3389/fnbot.2023.1084000>
- Su, Y., Chen, X., Zhou, T., Pretty, C., & Chase, G. (2022). Mixed reality-integrated 3D/2D vision mapping for intuitive teleoperation of mobile manipulator. *Robotics and Computer-Integrated Manufacturing*, *77*, 102332. <https://doi.org/10.1016/j.rcim.2022.102332>
- Sweller, J., van Merriënboer, J. J. G., & Paas, F. (2019). Cognitive architecture and instructional design: 20 years later. *Educational Psychology Review*, *31*(2), 261–292. <https://doi.org/10.1007/s10648-019-09465-5>
- Thees, M., Kapp, S., Altmeyer, K., Malone, S., Brünken, R., & Kuhn, J. (2021). Comparing two subjective rating scales assessing cognitive load during technology-enhanced STEM laboratory courses. *Frontiers in Education*. <https://doi.org/10.3389/educ.2021.705551>
- Wang, C. C., Wang, M., Sun, J., & Mojtahedi, M. (2021). A safety warning algorithm based on axis aligned bounding box method to prevent onsite accidents of mobile construction machineries. *Sensors*, *21*(21), 7075. <https://doi.org/10.3390/s21217075>
- Wang, T., Zheng, P., Li, S., & Wang, L. (2023). Multimodal human–robot interaction for human-centric smart manufacturing: A survey. *Advanced Intelligent Systems*. <https://doi.org/10.1002/aisy.202300359>
- Yang, W., Xiao, Q., & Zhang, Y. (2023). HA\$R^{2}\$Sbot: A human-centered augmented reality robot programming method with the awareness of cognitive load. *Journal of Intelligent Manufacturing*. <https://doi.org/10.1007/s10845-023-02096-2>
- Yashin, G. A., Trinitatova, D., Agishev, R. T., Ibrahimov, R., & Tsetserukou, D. (2019). AeroVr: Virtual Reality-based Teleoperation with Tactile Feedback for Aerial Manipulation. In *2019 19th International Conference on Advanced Robotics (ICAR)* (pp. 767–772). IEEE. <https://doi.org/10.1109/ICAR46387.2019.8981574>
- Yu, M., Lin, Y., Schmidt, D., Wang, X., & Wang, Y. (2014). Human–robot interaction based on gaze gestures for the drone teleoperation. *Journal of Eye Movement Research*. <https://doi.org/10.16910/jemr.7.4.4>
- Zhao, C., Du, S., Lv, J., Deng, Y., & Li, G. (2023a). A novel parallel classification network for classifying three-dimensional surface with point cloud data. *Journal of Intelligent Manufacturing*, *34*(2), 515–527. <https://doi.org/10.1007/s10845-021-01802-2>
- Zhao, C., Lui, C. F., Du, S., Wang, D., & Shao, Y. (2023b). An earth mover's distance based multivariate generalized likelihood ratio control chart for effective monitoring of 3D point cloud surface. *Computers & Industrial Engineering*, *175*, 108911. <https://doi.org/10.1016/j.cie.2022.108911>
- Zhao, C., Lv, J., & Du, S. (2022). Geometrical deviation modeling and monitoring of 3D surface based on multi-output Gaussian process. *Measurement*, *199*, 111569. <https://doi.org/10.1016/j.measurement.2022.111569>
- Zhou, T., Zhu, Q., & Du, J. (2020). Intuitive robot teleoperation for civil engineering operations with virtual reality and deep learning scene reconstruction. *Advanced Engineering Informatics*, *46*, 101170. <https://doi.org/10.1016/j.aei.2020.101170>
- Zhou, Y., Du, S., Liu, M., & Shen, X. (2024). Machine-fixtured-pallet resources constrained flexible job shop scheduling considering loading and unloading times under pallet automation system. *Journal of Manufacturing Systems*, *73*, 143–158. <https://doi.org/10.1016/j.jmsy.2024.01.010>
- Zhu, Y., Jiang, B., Chen, Q., Aoyama, T., & Hasegawa, Y. (2023). A shared control framework for enhanced grasping performance in teleoperation. *IEEE Access*, *11*, 69204–69215. <https://doi.org/10.1109/ACCESS.2023.3292410>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.